



An-Najah National University

Faculty of Engineering and Information Technology Department of Computer Science

Data Mining — Final Practical Project

" Smart Housing Data Mining System "

Submitted to: Dr. Ahmad Abu Alia

Submitted by — Group 12:

Mayar Basheer , Nancy Al-Jabareen , Hasan Jitan

Academic Year: 2025/2026

1. Group Information

Group Number : **12.**

Group Leader: **Mayar Basheer**

Project Title: **Smart Housing Data Mining System**

No.	Group Member Full Name
1	Mayar Basheer
2	Nancy Al-Jabareen
3	Hassan Jitan

2. Contribution and Main Tasks:

Member Name	Contribution Percentage	Main Tasks (top 5)
Mayar Basheer		1. Proposed the project idea and defined the problem.
		2. Collected and selected the apartment rental dataset.
		3. Prepared and cleaned the dataset for analysis.
		4. Reviewed, tested, and fixed the integrated project code.
		5. Developed the Streamlit interactive interface + Demo Video preparation
Hasan Jitan		1. Implemented K-Means clustering
		2. Applied PCA for dimensionality reduction and data visualization
		3. Helped review the project results and verify outputs
		4. Assisted in preparing presentation slides

Nancy Al-Jabareen		1. Implemented the content-based recommendation system using the clustered apartment dataset.
		2. Defined user preferences and built a similarity-based scoring system to rank apartment listings
		3. Generated the top recommended apartments and explained the recommendation results.
		4. Applied Isolation Forest to detect unusual apartment listings using selected numerical features.
		5. Created anomaly summaries and saved the recommendation and anomaly detection output files.

3. Repository, Demo, and Presentation Links

GitHub Repository URL:

<https://github.com/MayarBasheer/Data-Mining--Final-Project-.git>

Demo Video URL: *Add the demo video to your GitHub repository. Maximum duration: 90 seconds.*

<https://drive.google.com/file/d/1DUx8-igyTHAHCTAm0hIA7T0PvI7Saktt/view?usp=sharing>

Presentation File URL: *Add the presentation as a PDF file to your GitHub repository. Presentation duration: 6 minutes; one extra minute is allowed only for a group of four members.*

4. Simple Abstract:

Your abstract should simply answer these questions:

- What problem does your project solve?
- Why is this problem important?
- What data mining techniques did you use?
- What does your system/application do for the user?
- What are the main results or useful outputs?

Abstract:

This project presents a smart housing analysis and recommendation system for apartment rental listings. Finding suitable apartments can be difficult because users must compare many listings with different prices, locations, and apartment features. This problem is important because users often spend a lot of time searching for apartments that match their preferences and budget. To solve this problem, we used several unsupervised data mining techniques including K-Means clustering, PCA dimensionality reduction, Association Rule Mining using Apriori, anomaly detection using Isolation Forest, and a similarity-based recommendation system.. The application allows users to search apartments, receive personalized recommendations, explore apartment groups, and detect unusual listings through an interactive Streamlit web interface. The main outputs of the system include apartment recommendations, cluster analysis, visual insights, anomaly detection results, and interactive filtering tools that help users better understand the apartment market and find apartments more efficiently.

5. Data Mining Techniques Used:

Technique	Used? Yes/No	Algorithm / Method	Purpose in the Project
Clustering	Yes	K-Means	Group similar apartments
Dimensionality Reduction	Yes	PCA	Reduce feature complexity and visualize patterns
Recommender Systems	Yes	Content-based filtering	Recommend apartments based on user preferences
Time Series Analysis	No	–	–
Network Analysis	No	–	–
Anomaly Detection	Yes	Isolation Forest	Detect unusual apartment listings
Association Rule Mining	Yes	Apriori algorithm	Discover relationships between apartment amenities and listing features.

6. AI Generation Usage Declaration :

AI tools were used only for support, such as debugging, code review, English improvement, and report organization. The project idea, implementation decisions, testing, and final integration were completed by the group members.

AI Tool / Model	Purpose	Approx. Usage Ratio (%)
Claude	Code review, debugging, markdown writing	25%
ChatGPT	English writing improvement and code suggestions	10%

Total estimated AI-assisted contribution ratio: 20% of the whole project.

7. Limitations:

The recommendation system is content-based and does not learn from user behavior or feedback, which means it cannot improve over time based on user interactions.

Some listings have missing values in amenities and pets_allowed columns, which were replaced with "Unknown". This may affect the quality of Association Rule Mining results.

The Isolation Forest anomaly detection is set to flag 3% of listings as anomalies. This threshold was chosen manually and may not be optimal for all cases.

The dataset does not include rental price changes over time, so the system cannot analyze or predict price trends.

8. Future Work:

Expand the dataset to include more recent listings and international markets.

Implement collaborative filtering to improve recommendation quality using user interaction data.

Add a map-based visualization to display apartment locations geographically.

Deploy the application online so users can access it without local installation.

Include time series analysis to track rental price trends over time.

9. Special Contribution:

The strongest feature of this project is the combination of multiple data mining techniques integrated into a single user-friendly web application. Unlike typical data mining notebooks, our system translates complex analytical results into clear, practical insights for everyday users. The recommendation system uses normalized similarity scoring to fairly compare apartments across different price ranges and sizes. The Association Rule Mining page includes both a user-friendly Amenity Explorer and a technical view, making the system accessible to both regular users and data professionals. Additionally, the apartment comparison feature and market overview dashboard provide practical value that goes beyond standard data analysis.